

# CONSULTIQ

## AI-Assisted CV Parsing - Technical Approach for Demo 2

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| Document Purpose | Explain the CV parsing approach for Demo 2, the technology choices behind it, and how the approach will evolve in future phases. |
| Audience         | BBD Technical Stakeholders (Rudolph Esterhuysen, Tessa Engelbrecht, Xadrian Van Heerden, Ryan Trickett)                          |
| Date             | June 2026                                                                                                                        |

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### 1. Executive Summary

ConsultIQ is introducing CV parsing as part of the consultant profile creation flow. When a Consultant Manager uploads a consultant's CV, the system will automatically extract structured information - contact details, work experience, skills, and certifications - and pre-populate the profile creation form. The Consultant Manager then reviews and confirms the extracted data before the profile is saved.

This document explains the technical approach taken for Demo 2, the reasoning behind each technology choice, what is intentionally out of scope for this phase, and how the feature is expected to evolve afterward.

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### 2. How the Feature Works

CV parsing happens in four distinct stages. Each stage has a single, focused responsibility, which keeps the system easier to test, debug, and extend independently.

#### 2.1 Upload and Storage

The Consultant Manager uploads a CV file (PDF or Word document). The file is stored securely in a private AWS S3 bucket - not in our application database - and a reference to its location is recorded against the consultant's profile.

#### 2.2 Text Extraction

The system reads the actual words out of the uploaded file. Word documents and standard PDFs (i.e. ones created digitally, such as an exported Word CV) contain embedded text that can be extracted directly and with perfect accuracy. If a PDF turns out to be a scanned image with no embedded text, the system falls back to Optical Character Recognition (OCR) to read the text from the image.

## 2.3 Structured Parsing

The extracted text is analysed to identify meaningful sections - work experience, skills, certifications, contact details - and structured data is derived from each: job titles, company names, employment dates, skill names, and inferred competency levels.

## 2.4 Human Review and Approval

Before any data is saved, the Consultant Manager sees exactly what was extracted, with confidence indicators showing which fields the system is confident about and which may need manual correction. No consultant profile is ever created automatically without explicit human review and approval.

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## 3. Technology Choices and Rationale

For Demo 2, ConsultIQ's CV parsing does not call any external AI or cloud parsing APIs. This is a deliberate constraint for this phase, and it has shaped each technology decision below.

### 3.1 Text Extraction Libraries

| File Type           | Library Used | Why                                                                                                                                                                       |
|---------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PDF (digital)       | pdf-parse    | Extracts embedded text directly from PDF files with full accuracy and no external dependency.                                                                             |
| Word (.docx)        | mammoth      | Purpose-built for reading text content out of Word documents reliably.                                                                                                    |
| PDF (scanned image) | tesseract.js | Pure JavaScript OCR engine — runs entirely on our own server with no external API calls or per-document cost. Used only as a fallback when a PDF has no extractable text. |

### 3.2 The Parsing Logic Itself

Turning extracted text into structured profile data (recognising what is a skill, what is a job title, what is a date range) is implemented as custom rule-based logic - built using pattern matching and a curated dictionary of known skills and CV terminology, rather than a third-party parsing service or AI model.

This approach was chosen specifically because Demo 2 does not permit external API integrations for this feature. The benefits of this approach for the current phase are:

- **No ongoing cost per CV processed - everything runs on infrastructure we already control**
- **Fully explainable - for any extracted field, we can show exactly which pattern or rule produced it**

- **No dependency on third-party service availability or rate limits**
- **Full control over data handling, which simplifies POPIA compliance for this phase**

The tradeoff is that rule-based parsing requires a maintained library of patterns and known skill terms, and is less adaptable to highly unconventional CV layouts than a learning-based approach would be. This tradeoff is considered acceptable for Demo 2 and is mitigated by the mandatory human review step described in Section 2.4.

### 3.3 Background Processing

Because reading and analysing a CV takes a few seconds, this work happens in the background rather than making the Consultant Manager wait on a frozen screen. ConsultIQ uses BullMQ, a job queue system backed by Redis, to process uploads asynchronously: the file is queued for text extraction, then for parsing, and the Consultant Manager is notified once the result is ready for review.

### 3.4 File Storage

CV files are stored in AWS S3 in the af-south-1 (Cape Town) region, consistent with the rest of ConsultIQ's infrastructure and POPIA's requirement that personal information of South African residents remain within South Africa where reasonably possible. Files are never made public - they can only be retrieved through time-limited, secure links generated on demand.

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## 4. What Is In Scope for Demo 2

- **Upload of PDF and Word (.docx) CV files**
- **Automatic extraction of contact details, work experience, skills, and certifications**
- **Confidence scoring per extracted field**
- **Mandatory human review and correction before profile creation**
- **Fallback to manual profile entry at any point if a CV is unavailable or parsing is not confident**

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## 5. What Is Intentionally Out of Scope for Demo 2

The following were deliberately excluded from this phase. Each is a candidate for a future phase once there is time to implement it to the same standard of quality as the rest of the system.

### 5.1 Image-Based CVs (Photos, JPG/PNG)

Supporting photographed or image-based CVs requires a meaningfully different and less reliable processing pipeline than text-based documents - including image preprocessing

(deskewing, contrast correction, resolution handling) before OCR can even be attempted, and materially lower accuracy even with that preprocessing in place.

Given that the overwhelming majority of consultants have access to a digital copy of their CV (a Word document or a digitally generated PDF), this was assessed as a low-value, high-complexity addition for this phase. Manual profile entry remains available as a fallback for any consultant who only has a photographed CV.

## 5.2 AI/ML-Based Parsing

Using a third-party AI service (such as a large language model) to interpret CV content was not used in this phase, in line with the constraint that this feature should not rely on external API integrations for Demo 2. The rule-based approach described in Section 3.2 was used instead.

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## 6. Quality and Compliance Safeguards

- **Human-in-the-loop review:** no profile is created without explicit Consultant Manager confirmation of the extracted data
- **Confidence indicators:** fields the system is less certain about are visibly flagged for closer review
- **Graceful failure:** if a file cannot be processed (e.g. corrupted, unsupported format, unreadable scan), the system clearly communicates this and offers manual entry as an alternative - it never silently produces incorrect data
- **Data protection:** CV files are stored privately, accessed only through expiring secure links, and the same role-based access controls used elsewhere in ConsultIQ apply to this feature

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## 7. How This Will Evolve Beyond Demo 2

The current implementation has been deliberately structured so that the parsing logic is isolated behind a clear internal interface. This means the underlying parsing approach can be replaced or augmented later without requiring changes to the upload flow, the review screen, or how the rest of ConsultIQ consumes the resulting data.

### 7.1 AI-Based Parsing

Once external API integrations are permitted, the rule-based parsing engine can be replaced with an AI-based extraction service, which would likely improve accuracy on CVs with unconventional formatting and reduce the manual correction burden on Consultant Managers.

### 7.2 Image-Based CV Support

Support for photographed or scanned-image CVs can be added as a dedicated phase of work, including proper image preprocessing and accuracy testing against a representative sample of real submissions, rather than being added as a quick addition to the current pipeline.

### **7.3 Expanded Skills and Pattern Coverage**

The skills dictionary and section-recognition patterns will continue to be refined based on real-world CVs processed during testing and early use, improving extraction accuracy over time.

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## **8. Summary**

For Demo 2, ConsultIQ implements CV parsing using self-contained, rule-based text processing - no external AI services or paid APIs - covering PDF and Word document uploads, with Optical Character Recognition as a fallback for scanned PDFs. Every extracted result passes through mandatory human review before becoming part of a consultant's profile. This approach satisfies the current phase's constraints while keeping the system architected so that more advanced parsing capabilities can be introduced cleanly in future phases.